

Question D.4 — Decision Protocol and Sensitivity to Censoring

This section fixes the rule by which a coefficient of Q D is called discernible, reports censoring by amplitude rather than in aggregate, and settles whether censoring drives any result. Figures come from the campaign of 2 000 runs (20 amplitudes $\times$ 100 seeds, $H = 2\,000$, $M = 10$ trees in the adaptive random forest, ARF) and are reproduced by `scripts/chiffres_QD.py` and `scripts/derives_QD.py`. Every confidence interval (CI) on a τ_b is a bootstrap interval over the seeds of the stratum, 2 000 resamples, quantiles 2.5% and 97.5%; every interval on a fraction is a Wilson 95% interval.

1 A Reservation on the Status of the Criterion

The protocol asks for a decision criterion written down before any correlation is inspected. That claim cannot honestly be made here, so the reservation is stated rather than glossed.

The pre-registered document of this work fixes the criteria of a *benchmark*: the tests A1, A2, B1, B2, C1 and D1 to D3 that qualify candidate replacements for τ_{ARF} against an analytic reference. It contains neither the indicator evaluated here, nor its comparators, nor the decision domain $\Delta e \geq 0.10$, nor the threshold on τ_b , nor the rule on confidence-interval bounds. The columns `dans_domaine_decision`, `seuil_detectabilite` and `discernable` are produced by the same analysis pass as the coefficients they qualify. Anteriority is therefore not claimed for the criterion of Q D. It is fixed here, applied uniformly, and no cell is reclassified afterwards.

2 Two Incompatible Rules, and the One That Governs

Two rules for calling an association discernible coexist in this work.

Threshold rule. $|\tau_b| > 0.13295637$, the value in the `seuil_detectabilite` column, constant across the table.

Interval rule. The bootstrap confidence interval excludes zero. A τ_b whose interval contains zero does not count.

They disagree on exactly three cells out of the 108 in the decision domain.

Δe	comparator	τ_b	CI	threshold rule	interval rule
0.287	$\tau_{\text{swap}}(75\%)$	+0.1363	$[-0.0068; +0.2704]$	discernible	not
0.287	$S_{\text{max}}(H)$	+0.1308	$[+0.0027; +0.2519]$	not	discernible
0.391	$\tau_{\text{swap}}(75\%)$	+0.1240	$[+0.0010; +0.2505]$	not	discernible

Table 1: The three cells on which the two rules disagree.

The interval rule governs. Three reasons, of unequal weight. A fixed threshold on $|\tau_b|$ ignores the precision of the estimate, treating a coefficient of 0.13 measured on 100 seeds with a wide interval exactly like one measured tightly, whereas the interval rule uses what the bootstrap actually provides. The threshold is also a function of n alone, so it says nothing about the tie structure of the particular pair being tested, and Q D.2 established that this structure differs systematically between the $\tau_{\text{swap}}(q)$ comparators and the never-censored $A(H)$ and $S_{\text{max}}(H)$. Finally, the pre-registered benchmark already uses the interval rule for test B1, so adopting it keeps one rule across the project instead of two.

The counts of Q D.3 are those this rule gives: $A(H)$ discernible at 0 of 18 amplitudes, unchanged since the two rules agree on every $A(H)$ cell, and $S_{\text{max}}(H)$ at **15** of 18. The `discernable` column

implements the threshold rule and would give 14: the difference is the single cell $\Delta e = 0.287$, whose interval $[+0.0027; +0.2519]$ excludes zero while its τ_b of 0.1308 falls just under the threshold. The count reported throughout Q D is 15, and the column is not used as the criterion. None of the three cells of Table 1 is quoted individually as a result.

3 Why Censored Runs May Be Ranked Last

The horizon $H = 2000$ is identical for every run, so a duration not reached within it genuinely exceeds every observed duration and censored runs may be placed at the last rank, tied among themselves. Kendall’s τ_b then removes those pairs from the count on the side where the tie occurs, as Q D.2 sets out.

The argument depends entirely on the horizon being common. If H varied from run to run, a censored value would only be known to exceed *its own* horizon: a run censored at $H = 500$ and an observed run of 1 200 steps could not be ordered at all, and no imputation to a shared last rank would be licit. Under variable horizons the correct tool is a censoring-aware estimator such as Harrell’s C index. The convention used here rests on a property of this experimental design and does not transfer.

4 Censoring by Amplitude

Aggregate censoring rates conceal what matters.

Quantity	overall	worst on grid (at Δe)	worst in domain (at Δe)
τ_{ARF}	0.0000	0.0000	0.0000
$A(H), S_{\text{max}}(H)$	0.0000	0.0000	0.0000
$\tau_{\text{swap}}(25\%)$	0.0005	0.0100 (0.141)	0.0100 (0.141)
$\tau_{\text{swap}}(50\%)$	0.0110	0.1200 (0.085)	0.0600 (0.141)
$\tau_{\text{swap}}(75\%)$	0.0910	0.7800 (0.085)	0.5400 (0.141)
$\tau_{\text{det}}, \lambda = 8$	0.0470	0.9000 (0.028)	0.0000 (all 18)
$\tau_{\text{det}}, \lambda = 25$	0.6110	1.0000 (0.028)	0.9900 (0.492, 0.494, 0.496)
$\tau_{\text{det}}, \lambda = 50$	0.9995	1.0000 (every Δe but 0.194)	1.0000 (every Δe but 0.194)

Table 2: Censoring overall, at its worst amplitude on the full grid, and at its worst amplitude inside the decision domain. The columns differ because the two amplitudes below $\Delta e = 0.10$ carry most of the censoring and lie outside the domain on which the criterion is declared. Baseline estimated over 3 000 pre-drift steps. At $\lambda = 50$ the single run that alarms is at $\Delta e = 0.194$.

$\tau_{\text{swap}}(75\%)$ fails the group’s own criterion. The criterion adopted for the benchmark requires censoring of at most 5% over the declared domain. Inside the decision domain, $\tau_{\text{swap}}(75\%)$ reaches 54% at $\Delta e = 0.141$, exceeding the criterion by a factor of 10.8. Its worst value on the full grid is 78%, at $\Delta e = 0.085$, but that amplitude lies outside the decision domain and the figure is quoted as a grid measurement rather than as a breach of an in-domain criterion. Its coefficients are published in Q D.3 because suppressing them would hide the comparator that most stresses the method, but they are published with this failure stated: at $\Delta e = 0.141$, the lowest amplitude of the decision domain, the τ_b of $\tau_{\text{swap}}(75\%)$ rests on a minority of observed runs and is not quoted as evidence.

No median delay beyond 50% censoring. At $\lambda = 25$ and $\lambda = 50$ no median detection delay is reported anywhere in Q D, and at $\lambda = 8$ none is reported at $\Delta e = 0.028$ where censoring reaches 90%. A median computed on the surviving runs excludes the slowest ones by construction and is not a median of the quantity of interest.

5 Does Censoring Drive the Results?

Each coefficient was recomputed twice, once with censored runs imputed to the last rank and once on complete cases only, then compared cell by cell.

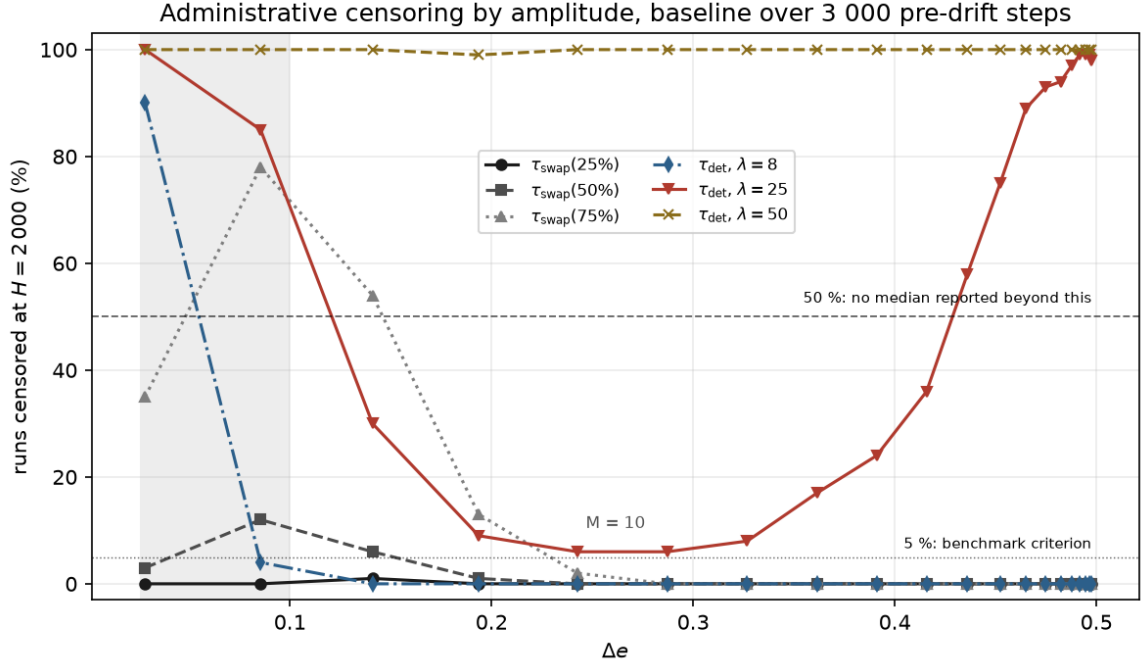


Figure 1: Censoring at $H = 2000$ of every censorable duration, by amplitude, one series per duration, with the 50% line above which no median is reported and the 5% line of the benchmark criterion; baseline over 3000 pre-drift steps. What to see: $\tau_{\text{swap}}(75\%)$ peaks at 78% at $\Delta e = 0.085$ and is still above 50% at $\Delta e = 0.141$, the lowest amplitude of the decision domain, but only 35% at the very bottom of the grid, and it falls under 5% from $\Delta e = 0.243$; τ_{det} at $\lambda = 25$ is U-shaped, below 10% only between $\Delta e = 0.19$ and 0.33 ; at $\lambda = 50$ it never leaves 99%. Tables `QCD_correlations_stratifiees_full` and `QCD_socle_deux_fenetres`.

Comparator	max imputed – complete cases
$\tau_{\text{swap}}(10\%), A(H), S_{\max}(H)$	0.000000
$\tau_{\text{swap}}(25\%)$	0.000228
$\tau_{\text{swap}}(50\%)$	0.030591
$\tau_{\text{swap}}(75\%)$	0.040259

Table 3: Sensitivity of τ_b to the treatment of censored runs.

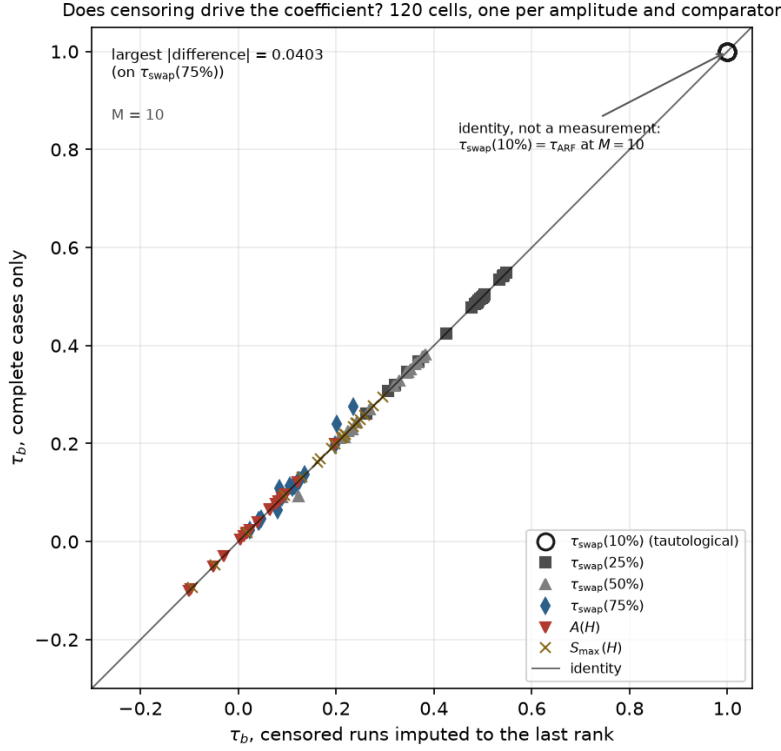


Figure 2: The 120 cells of Table 3 one by one: τ_b on complete cases against τ_b with censored runs imputed to the last rank, one marker per comparator, with the identity line. The hollow circle at (1, 1) is $\tau_{\text{swap}}(10\%)$, an identity at $M = 10$ and not a measurement. Table `QCD_cas_complets_full`.

Answer: no. For the three never-censored comparators the two computations agree exactly, to 0.000000, which is what they must do and serves as a control on the implementation. For the two materially censored comparators the largest disagreement anywhere in the table is 0.0403, on $\tau_{\text{swap}}(75\%)$, an order of magnitude below the coefficients that carry the conclusions of Q D.3 and well inside their confidence intervals. **Censoring does not drive any result of question D.** The imputation convention could be replaced by complete-case analysis without changing a verdict.

One correction to an earlier internal note: the claim that imputed and complete-case values “agree to three decimal places” is too strong. It holds for $\tau_{\text{swap}}(25\%)$ and for the never-censored comparators; for $\tau_{\text{swap}}(50\%)$ and $\tau_{\text{swap}}(75\%)$ the largest differences are 0.0306 and 0.0403, which differ at the second decimal place, and it is these two numbers that are quoted.

6 The Blind Spot, Quantified

λ	runs that alarm	silent	$\tau_{\text{det}} < \tau_{\text{ARF}}$
8	1 906/2 000	4.70%	95.44%
25	778/2 000	61.10%	6.56%
50	1/2 000	99.95%	0.00%

Table 4: Alarm behaviour by threshold. The last column is computed on uncensored runs only, and is the race of question A read from this campaign.

At $\lambda = 50$ a single run out of 2000 ever alarms. The methodological consequence is why the campaign fixes no λ : with 99.95% censoring, no correlation involving τ_{det} is computable at that threshold. Had a threshold been frozen inside the simulation loop, as an earlier prototype did, the resulting table would carry a τ_{det} column empty in all but one row.

The last column sharpens the picture, with one qualification. Across the grid the detector wins the race in 95.44% of uncensored runs at $\lambda = 8$ and 6.56% at $\lambda = 25$, so the transition from “the detector is never late” to “the detector never fires” happens entirely within a narrow band of the tuning parameter.

But that rate is not stationary in amplitude: at $\lambda = 8$ it is 0% at $\Delta e = 0.028$, 64.6% at $\Delta e = 0.085$ (62 of 96 uncensored runs), and 88% to 99% above (Q D.1, Table 2). The blind spot depends on the pair $(\lambda, \Delta e)$, and lowering λ does not remove it at the bottom of the grid.

7 Effect of the Baseline Window

Since $\hat{p}_0$ enters S_t directly, every censoring rate above depends on the window used to estimate it. Both windows required by the protocol are reported in Q D.1: mean censoring of τ_{det} moves from 0.0405 to 0.0470 at $\lambda = 8$, from 0.6045 to 0.6110 at $\lambda = 25$, and from 0.9990 to 0.9995 at $\lambda = 50$ as the window goes from 1 000 to 3 000 steps. The shifts stay below one point and change no verdict of this section, but the direction is systematic: a longer window gives a slightly higher baseline, hence slightly less accumulated evidence and slightly fewer alarms.

8 Not Addressed

No result of Q D is established as a function of M , since the instrumented campaign exists at $M = 10$ only. No result involves τ_{det} at $\lambda = 25$ or $\lambda = 50$ as a correlated variable, for the censoring reason given above. Harrell's C index was not used, because the common-horizon argument makes rank imputation licit here and the complete-case comparison shows the choice is immaterial; under a variable horizon that reasoning would fail and the C index would be required.